

2024

The Itaú logo is displayed in white lowercase letters on a large, rounded orange background. The letters are bold and modern, with a slight shadow effect.

itaú



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Itaú Improves Speed to Market and Product Development with ML Solutions Using Amazon Web Services



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Learn how Itaú, Latin America's largest bank, improved speed to market for ML models using Amazon SageMaker Studio.

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6 months to 3–5 days

decrease in deployment time

Improved speed to market

leading to improved customer experience

Increased staff productivity

with standardization

Improved solution integration

for ML data scientists

+3,200 users

on Amazon SageMaker Studio

Overview

[Itaú Unibanco](#) (Itaú), the largest private-sector bank in Brazil, needed to improve the speed, flexibility, and scalability of its machine learning (ML) infrastructure for its more than 3,200 ML users. The bank's on-premises infrastructure required ordering servers and completing configuration tasks before solutions were available to the data science team. This process took months and came with the high associated costs of purchasing servers and running and housing a data center.

In 2020, Itaú chose Amazon Web Services (AWS) as a strategic cloud provider and began renovating its infrastructure on AWS. To speed up ML processes for data scientists, Itaú used [Amazon SageMaker Studio](#), an integrated development environment that provides a single web-based visual interface to access purpose-built tools to perform all ML development steps. The company felt Amazon SageMaker Studio was the obvious choice for its solution. With its new solution, Itaú improved model development time from 6 months to 5 days, increased staff productivity with standardization, and reduced costs.



Opportunity | Using Amazon SageMaker Studio to Efficiently Democratize ML for Itaú

Itaú provides banking services to customers in Brazil, Latin America, and 18 other countries around the world. It has over 95,700 employees, about 15,000 of whom work in IT. Itaú's original infrastructure was entirely on premises, which led to high costs and slow development times. The on-premises infrastructure was also not scalable because it was limited by physical space and hardware. In the bank's data group, data scientists would have to wait up to 6 months for memory and resources to be made available, and the company had a deployment waiting list with over 100 ML models on it.

To overcome these challenges, Itaú decided to migrate a portion of its business to the cloud and chose to use AWS. "One of the reasons we chose to migrate from on premises to the cloud was a strategy to increase business competitiveness and efficiency at the same time," says Diego Nogare, ML engineering manager at Itaú.

Around 6 months after the start of the migration, Itaú chose [Amazon SageMaker](#)—a service to build, train, and deploy ML models for any use case with fully managed infrastructure, tools, and workflows—as a flexible, cloud-native ML solution. "We were transforming our software and data using AWS, and we needed a solution that worked perfectly on AWS," says Vitor Azeka, data science superintendent at Itaú. "Amazon SageMaker was the obvious choice." As of 2024, around 60 percent of the company's software and data is already modernized to run on cloud".

Solution | Decreasing Model Deployment Time from 6 Months to 5 Days Using AWS

Itaú has built a complete solution for its data scientists using AWS. First, data is gathered using [AWS Glue](#), a serverless data integration service that makes it easier to discover, prepare, move, and integrate data from multiple sources for analytics, ML, and application development. This data is then used to start experiments using Amazon SageMaker Studio. Itaú uses Amazon SageMaker Studio as a flexible development solution for its internal data scientists to experiment. Next, ML models are deployed using other Amazon SageMaker tools, such as Endpoints, Batch Transform, and Asynchronous Inference. The company monitors models using [Amazon CloudWatch](#), which collects and visualizes near-real-time logs, metrics, and event data in automated dashboards to streamline infrastructure and application maintenance. Using all these AWS services together, data scientists can achieve their needs.

Itaú delivered its first solution using Amazon SageMaker Studio as an integrated development environment in August 2021, and as of April 2023, it had more than 3,200 unique users for the AWS service, including around 350 data scientists.

Itaú no longer has a waiting list for deploying ML models. Using Amazon SageMaker Studio, the company has reduced deployment time from up to 6 months to 3–5 days in some cases. This reduced deployment time improves speed to market for the company. “When we use Amazon SageMaker Studio, we can run our pipeline and deliver the solution to our customers very quickly,” says Nogare. “We can thus improve customer experience.” Itaú is also saving on costs compared with its old on-premises infrastructure.

Since November 2021, Itaú has had weekly meetings with the AWS team to discuss architecture, security, and its road map. “AWS support was very important to achieve the results we have today,” says Nogare. “Any time we saw a problem with our solution or governance needs, the AWS team supported us through that.” Some of the governance needs are met using Amazon SageMaker Studio. When the company runs pipelines to provide Amazon SageMaker Studio to users, issues of governance and security are already resolved.

The standardization of its solution means Itaú can more easily onboard new employees and move data scientists from one department to another. Updating is easier since everything is virtual, and the company no longer needs to rely on physical machines. Using AWS, the pipelines for data scientists are integrated, so ML models are deployed and monitored in the same data pipeline. This further improves efficiency for data scientists.

“At the end of the day, we can deliver faster,” says Rodrigo Fernandes Mello, distinguished data scientist at Itaú. “We have improved standardization and integration, and we can use AWS to continue improving.”

Outcome | Standardizing for Efficiency Using AWS

Itaú is looking to continue improving its standardization. The next step for internal standardization for its data scientists involves having more employees using its IARA solution, which is based on AWS and services, including Amazon SageMaker Studio. Itaú will continue to develop its pipeline by using Amazon SageMaker, such as [Amazon SageMaker Pipelines](#), which is used to create, automate, and monitor workflows at scale. Itaú is running tests to bring more standardization to its pipeline using AWS tools.



“This project brought a lot of efficiency to the data scientist team,” says Azeka. “Using Amazon SageMaker Studio, we can test new things while publishing others, and we can discuss state-of-the-art solutions using large language models. This makes our data scientists proud to work at Itaú.”

About Itaú Unibanco

Itaú is the largest private-sector bank in Brazil and provides whole banking, which includes corporate banking, investment banking, and retail banking investment. The company was formed through the merger of Banco Itaú and Unibanco in 2008.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

Amazon SageMaker Studio

Amazon SageMaker Studio offers a wide choice of purpose-built tools to perform all machine learning (ML) development steps, from preparing data to building, training, deploying, and managing your ML models.

[Learn more »](#)

AWS Glue

Preparing your data to obtain quality results is the first step in an analytics or ML project. AWS Glue is a serverless data integration service that makes data preparation simpler, faster, and cheaper.

[Learn more »](#)

Amazon CloudWatch



Amazon CloudWatch is a service that monitors applications, responds to performance changes, optimizes resource use, and provides insights into operational health.

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Itaú on AWS

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FRANCE



AWS Announces Amazon DataZone

At AWS re:Invent, Amazon Web Services Inc. (AWS), an Amazon.com, Inc. company (NASDAQ: AMZN), today announced Amazon DataZone, a new data management service that makes it

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faster and easier for customers to catalog, discover, share, and govern data stored across AWS, on-premises, and third-party sources.

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partnered with AWS to improve the resilience and availability of its banking applications for its customers as part of its digital transformation.

2023

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